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1 $A=\begin{pmatrix} a & b \\ 0 & d \end{pmatrix}$

① $a=d$ $A=\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ or $A=\begin{pmatrix} 0 & k \\ \frac{1}{k} & 0 \end{pmatrix}$ $k\neq 0$.

② $a=-d$ $A=\begin{pmatrix} k_1 & k_2 \\ \frac{1-k_1^2}{k_2} & -k_1 \end{pmatrix}$.

2. (1) $\left[\begin{array}{ccccc|c} 1 & -1 & 0 & 2 & 1 & 0 \\ 3 & -3 & 0 & 7 & 0 & 0 \\ 1 & -1 & 2 & 3 & 2 & 0 \\ 2 & 2 & 2 & 7 & -3 & 0 \end{array}\right] \to \left[\begin{array}{ccccc|c} 1 & -1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & 1 & -3 & 0 \\ 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 2 & 3 & -5 & 0 \end{array}\right] \to \left[\begin{array}{ccccc|c} 1 & -1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 2 & -6 & 0 \\ 0 & 0 & 0 & 1 & -3 & 0 \end{array}\right] \to \left[\begin{array}{ccccc|c} 1 & -1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 2 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$

$\Rightarrow \begin{cases} x_1 = k_1 - 7k_2 \\ x_2 = k_1 \\ x_3 = -2k_2 \\ x_4 = 3k_2 \end{cases}$

(2) $\left[\begin{array}{ccc|c} 1 & 2 & 3 & 8 \\ 2 & 5 & 9 & 16 \\ 3 & -9 & -5 & 32 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & 3 & 8 \\ 0 & 1 & 3 & 0 \\ 0 & -10 & -14 & 8 \end{array}\right] \to \left[\begin{array}{ccc|c} 1 & 2 & 3 & 8 \\ 0 & 1 & 3 & 0 \\ 0 & 0 & 16 & 8 \end{array}\right]$

$\Rightarrow \begin{cases} x_1 = \frac{19}{2} \\ x_2 = -\frac{3}{2} \\ x_3 = \frac{1}{2} \end{cases}$

Q3. (1) $\begin{bmatrix} 3 & 2 & 0 & 1 & 0 \\ 1 & 3 & 0 & 0 & 1 \\ -2 & 4 & 0 & -1 & 2 \\ 4 & 4 & 0 & 1 & 1 \\ -6 & -9 & -1 & -2 & 0 \end{bmatrix}$

(2) $\begin{bmatrix} \frac{1}{2} & 0 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 & 0 \\ 0 & 0 & 0 & \frac{1}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{3} & -\frac{2}{3} \\ 0 & 0 & -1 & \frac{2}{3} & -\frac{1}{3} \end{bmatrix}$

Q4. $(A-2I)BA^T = I$.

$A-2I = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & 2 \\ 0 & 0 & -1 \end{bmatrix}$ is invertible.

$(A-2I)^{-1} = \begin{bmatrix} -1 & 0 & 0 \\ 0 & -1 & -2 \\ 0 & 0 & -1 \end{bmatrix}$

$(A^T)^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -2 & 1 \end{bmatrix}$

$B = (A-2I)^{-1}(A^T)^{-1}$

$= \begin{bmatrix} -1 & 0 & 0 \\ 0 & 3 & -2 \\ 0 & 2 & -1 \end{bmatrix}$

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MAT2040: Midterm Exam

Xiang Li, 220019035

Question 5

Matrix $A \in \mathbb{R}^{n\times n}$ is invertible, and matrix B is generated by interchaning the $i-$th and $j-th$ rows of A. *(10 points)*

1) Please check whether B is invertible? why?

2) and compute AB^{-1}.

Solution:

1) Denote E_{R_i,R_j} the row exchange matrix,

$$B = E_{R_i,R_j}A \tag{1}$$

Since E_{R_i,R_j} and A is invertible, B is invertible.

Specifically, $E^{-1}_{R_i,R_j} = E_{R_i,R_j}$ and $B^{-1} = A^{-1}E^{-1}_{R_i,R_j} = A^{-1}E_{R_i,R_j}$.

2)

$$AB^{-1} = E_{R_i,R_j} \tag{2}$$

Question 6

Please compute the determinant of the following n by n matrix: *(10 points)*

$$\begin{bmatrix} 2 & 1 & 0 & \cdots & 0 & 0 \\ 1 & 2 & 1 & \cdots & 0 & 0 \\ 0 & 1 & 2 & \cdots & 0 & 0 \\ \cdots & & & & & \\ 0 & 0 & 0 & \cdots & 2 & 1 \\ 0 & 0 & 0 & \cdots & 1 & 2 \end{bmatrix} \tag{3}$$

Solution:

Denote the matrix (3) as A:

$$det(A) = 2\begin{vmatrix} 2 & 1 & \cdots & 0 & 0 \\ 1 & 2 & \cdots & 0 & 0 \\ \cdots & & & & \\ 0 & 0 & \cdots & 2 & 1 \\ 0 & 0 & \cdots & 1 & 2 \end{vmatrix} - \begin{vmatrix} 1 & 1 & \cdots & 0 & 0 \\ 0 & 2 & \cdots & 0 & 0 \\ \cdots & & & & \\ 0 & 0 & \cdots & 2 & 1 \\ 0 & 0 & \cdots & 1 & 2 \end{vmatrix} \tag{4}$$

Denote $det(A)$ as D_n, then

$$D_n = 2D_{n-1} - D_{n-2} \tag{5}$$

$$D_n - D_{n-1} = D_{n-1} - D_{n-2} = D_2 - D_1 = 1 \tag{6}$$

$$D_n = D_{n-1} + 1 = n - 1 + D_1 = n + 1 \tag{7}$$

Question 7

Please compute the determinant of the following n by n matrix: *(10 points)*

$$\begin{bmatrix} 1 & 2 & 2 & \cdots & 2 & 2 \\ 2 & 2 & 2 & \cdots & 2 & 2 \\ 2 & 2 & 3 & \cdots & 2 & 2 \\ \cdots & & & & & \\ 2 & 2 & 2 & \cdots & n-1 & 2 \\ 2 & 2 & 2 & \cdots & 2 & n \end{bmatrix} \tag{8}$$

Solution: Take elementary row operation, i.e., let each row minus the second row (except the second row):

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$$\begin{bmatrix} -1 & 0 & 0 & \cdots & 0 & 0 \\ 2 & 2 & 2 & \cdots & 0 & 0 \\ 0 & 0 & 1 & \cdots & 0 & 0 \\ \cdots & & & & & \\ 0 & 0 & 0 & \cdots & n-3 & 0 \\ 0 & 0 & 0 & \cdots & 0 & n-2 \end{bmatrix} \tag{9}$$

Denote matrix (9) as A:

$$det(A) = -\begin{vmatrix} 2 & 2 & \cdots & 0 & 0 \\ 0 & 1 & \cdots & 0 & 0 \\ \cdots & & & & \\ 0 & 0 & \cdots & n-3 & 0 \\ 0 & 0 & \cdots & 0 & n-2 \end{vmatrix} \tag{10}$$

Notice that the above matrix is an upper triangular square matrix:

$$det(A) = -2(n-2)! \tag{11}$$

Question 8

Please compute the determinant of the following 4 by 4 matrix: (*10 points*)

$$\begin{bmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ 1+a_2b_1 & 1+a_2b_2 & 1+a_2b_3 & 1+a_2b_4 \\ 1+a_3b_1 & 1+a_3b_2 & 1+a_3b_3 & 1+a_3b_4 \\ 1+a_4b_1 & 1+a_4b_2 & 1+a_4b_3 & 1+a_4b_4 \end{bmatrix} \tag{12}$$

Solution: Take elementary row operation, i.e., let each row minus the first row (except the first row):

4

$$\begin{bmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ (a_2-a_1)b_1 & (a_2-a_1)b_2 & (a_2-a_1)b_3 & (a_2-a_1)b_4 \\ (a_3-a_1)b_1 & (a_3-a_1)b_2 & (a_3-a_1)b_3 & (a_3-a_1)b_4 \\ (a_4-a_1)b_1 & (a_4-a_1)b_2 & (a_4-a_1)b_3 & (a_4-a_1)b_4 \end{bmatrix} \tag{13}$$

Denote matrix (13) as matrix A:

$$det(A) = (a_2-a_1)(a_3-a_1)(a_4-a_1)\begin{vmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ b_1 & b_2 & b_3 & b_4 \\ b_1 & b_2 & b_3 & b_4 \\ b_1 & b_2 & b_3 & b_4 \end{vmatrix} = 0 \tag{14}$$

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Midterm Problem 1

- Please find all the Matrices $A \in \mathbb{R}^{2\times 2}$ which satisfy $A^2 = I_2$

Solution:

Suppose $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$, we have: $A^2 = \begin{bmatrix} a^2+bc & ab+bd \\ ac+cd & bc+d^2 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$.

then,

$$a^2 + bc = d^2 + bc = 1$$
$$b(a+d) = c(a+d) = 0$$

Case1: b = c = 0.

we have, $a^2 = d^2 = 1$, which leads to a $= d = \pm 1$.

$A = I_2$ or $A = -I_2$

Case2: $\neg(b = c = 0), a = -d$

we have $a^2 + bc = 1$, which gives us: $a = \pm\sqrt{1-bc}$, then

$$\mathrm{A} = \begin{bmatrix} \pm\sqrt{1-bc} & b \\ c & \mp\sqrt{1-bc} \end{bmatrix}, bc \le 1, b \in \mathbb{R}, c \in \mathbb{R}$$

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Problem 8 Solution

Solution: Take elementary row operation, i.e., let each row minus the first row (except the first row):

$$\begin{bmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ (a_2-a_1)b_1 & (a_2-a_1)b_2 & (a_2-a_1)b_3 & (a_2-a_1)b_4 \\ (a_3-a_1)b_1 & (a_3-a_1)b_2 & (a_3-a_1)b_3 & (a_3-a_1)b_4 \\ (a_4-a_1)b_1 & (a_4-a_1)b_2 & (a_4-a_1)b_3 & (a_4-a_1)b_4 \end{bmatrix}$$

Property 16.7 (**Determinant for an Elementary Matrix Multiply a Matrix**) Suppose that A is a square matrix of size n and E is an elementary matrix of size n, then

$$\det(EA) = \det(E)\det(A)$$

$$\det(AE) = \det(A)\det(E)$$

Denote matrix (13) as ~~matrix~~ A:

$$det(A) = (a_2-a_1)(a_3-a_1)(a_4-a_1)\begin{vmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ b_1 & b_2 & b_3 & b_4 \\ b_1 & b_2 & b_3 & b_4 \\ b_1 & b_2 & b_3 & b_4 \end{vmatrix} = 0$$

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Problem 8 Solution

Solution: Take elementary row operation, i.e., let each row minus the first row (except the first row):

$$\begin{bmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ (a_2-a_1)b_1 & (a_2-a_1)b_2 & (a_2-a_1)b_3 & (a_2-a_1)b_4 \\ (a_3-a_1)b_1 & (a_3-a_1)b_2 & (a_3-a_1)b_3 & (a_3-a_1)b_4 \\ (a_4-a_1)b_1 & (a_4-a_1)b_2 & (a_4-a_1)b_3 & (a_4-a_1)b_4 \end{bmatrix}$$

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Recording

Problem 8 Solution

Solution: Take elementary row operation, i.e., let each row minus the first row (except the first row):

$$\begin{bmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ (a_2-a_1)b_1 & (a_2-a_1)b_2 & (a_2-a_1)b_3 & (a_2-a_1)b_4 \\ (a_3-a_1)b_1 & (a_3-a_1)b_2 & (a_3-a_1)b_3 & (a_3-a_1)b_4 \\ (a_4-a_1)b_1 & (a_4-a_1)b_2 & (a_4-a_1)b_3 & (a_4-a_1)b_4 \end{bmatrix}$$

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$$\det(EA) = \det(E)\det(A)$$

$$\det(AE) = \det(A)\det(E)$$

Denote matrix (13) as matrix A:

$$det(A) = (a_2-a_1)(a_3-a_1)(a_4-a_1)\begin{vmatrix} 1+a_1b_1 & 1+a_1b_2 & 1+a_1b_3 & 1+a_1b_4 \\ b_1 & b_2 & b_3 & b_4 \\ b_1 & b_2 & b_3 & b_4 \\ b_1 & b_2 & b_3 & b_4 \end{vmatrix} = 0$$

Midterm Problems

Question 5 ..... *10 points*

Matrix $A \in \mathbb{R}^{n\times n}$ is invertible, and matrix B is generated by interchanging the i-th and j-th rows of A. (1) Please check whether B is invertible? why? (2) and compute AB^{-1}.

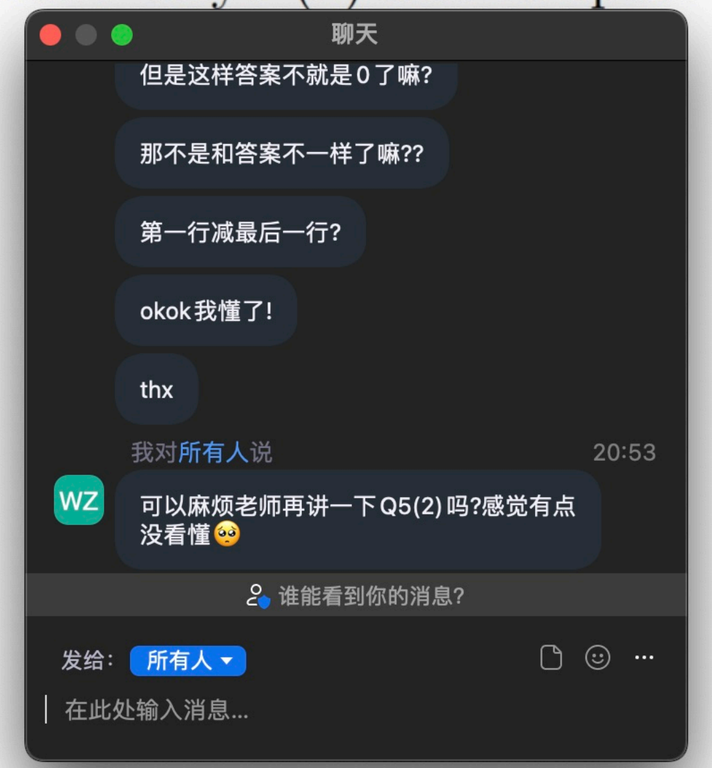
Solution:

1) Denote E_{R_i,R_j} the row exchange matrix,

$$B = E_{R_i,R_j}A$$

Since E_{R_i,R_j} and A is invertible, B is invertible.

Specifically, $E^{-1}_{R_i,R_j} = E_{R_i,R_j}$ and $B^{-1} = A^{-1}E^{-1}_{R_i,R_j} = A^{-1}E_{R_i,R_j}$.



Problem 5 Solution

(2)
Since $B = E_{R_iR_j}A$, $B^{-1} = A^{-1}E_{R_iR_j}^{-1} = A^{-1}E_{R_iR_j}$,
then

$$AB^{-1} = AA^{-1}E_{R_iR_j} = E_{R_iR_j}$$

Theorem 6.10 (**Elementary Matrices are Invertible and Their Inverse are also Elementary Matrices**)

(1) $E_{R_iR_j}^{-1} = E_{R_iR_j}$, corresponding to the reverse row operation 1: $R_i \leftrightarrow R_j$.

(2) $E_{\alpha R_i}^{-1} = E_{\frac{1}{\alpha}R_i}$ $(\alpha \neq 0)$, corresponding to the reverse row operation 2: $R_i \to \frac{1}{\alpha}R_i$.

(3) $E_{\beta R_i+R_j}^{-1} = E_{-\beta R_i+R_j}$, corresponding to the reverse row operation 3: $R_j \to -\beta R_i + R_j$.

